

The Implicit Bias of Steepest Descent with Mini-batch Stochastic Gradient



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Motivation

Modern optimizers such as SignSGD and Muon can be interpreted as normalized **steepest descent** under different norm-induced geometries.

Norm geometry	Steepest map	Optimizer family
Frobenius / ℓ_2	$\mathbf{G} / \ \mathbf{G}\ _2$	Normalized-SGD
Max / ℓ_∞	$\text{sign}(\mathbf{G})$	SignSGD / Signum
Spectral	$\mathbf{U}\mathbf{V}^\top$ (SVD)	Spectral-SGD / Muon

Full-batch implicit bias: In linear-classification, normalized steepest descent with entry-wise or Schatten-p norms maximizes the margin w.r.t. the corresponding norm.

Question: Does the **mini-batch stochastic steepest descent** have the same **max-margin implicit bias** as **full-batch steepest descent**?

Problem Setup

Training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in [k]$, Linear classifier $\mathbf{W} \in \mathbb{R}^{k \times d}$. Cross-Entropy/Exponential loss:

$$L(\mathbf{W}) = -\frac{1}{n} \sum_{i=1}^n \log \hat{p}(y_i | \mathbf{x}_i) = -\frac{1}{n} \sum_{i=1}^n \log \mathbb{S}_{y_i}(\mathbf{W}\mathbf{x}_i).$$

For **any** entry-wise or Schatten-p norm, the **norm-induced max margin** is:

$$\gamma := \max_{\|\mathbf{W}\| \leq 1} \min_{i \in [n], c \neq y_i} (\mathbf{e}_{y_i} - \mathbf{e}_c)^\top \mathbf{W}\mathbf{x}_i.$$

We assume **linear separability**, **bounded data**, **random-reshuffling scheme** and **decaying stepsizes** $\eta_t = \Theta(\frac{1}{t^a})$

Optimization Algorithms. At each iteration, the parameters are updated as: $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta_t \Delta_t$,

Given any entry-wise or Schatten-p norm, the steepest direction is given by:

$$\Delta_t := \phi_{\|\cdot\|}(\mathbf{G}_t) := \arg \max_{\|\Delta\| \leq 1} \langle \mathbf{G}_t, \Delta \rangle.$$

Signals include **gradient**, **momentum**, **SVRG-style Variance reduction update** as well as its momentum version.

$$\mathbf{G}_t \in \{\nabla \mathcal{L}_{B_t}(\mathbf{W}_t), \mathbf{M}_t, \mathbf{V}_t, \mathbf{M}_t^V\}$$

Main Theoretical Results

Let b be batch size, $m = n/b$, $\eta_t = c \cdot t^{-a}$ where $a \in (0, 1]$

NSD (without momentum)

Sufficient condition for convergence and successful classification:

$$\rho := \gamma - 4(\frac{n}{b} - 1)R > 0 \Rightarrow b = \Omega(n \frac{R}{\gamma + R}), \xrightarrow{\text{Forcing}} \mathbf{b} = \mathbf{n} \text{ (Full-batch)}$$

Counter-example (where SignSGD Fails classification with $m=2$) :

$$(\mathbf{x}_1, y_1) = ((1, -\varepsilon), 1), \quad (\mathbf{x}_2, y_2) = ((\varepsilon, -1), 2).$$

► **Key Insight:** **Without momentum**, worst-case convergence for any norm (general algorithm) and successful classification can **only be guaranteed with full-batch gradient**. (So stabilizing mechanism is important for vanilla-NSD!)

NSDM (with momentum)

If the **positive effective margin condition** (sufficient condition) satisfies, where

$$\text{approximate max-margin } \rho := \gamma - 2(1 - \beta_1)m(m^2 - 1)R > 0, \Rightarrow b = \Omega(n(\frac{(1 - \beta_1)R}{\gamma})^{1/3}).$$

Set $a = 1/2$, then **NSD with momentum converges to an Effective Approximate max-margin solution as full-batch implicit bias**.

$$\rho - \text{margin}(\mathbf{W}_t) = O\left(\frac{\frac{m}{1 - \beta_1}(n + \log t)}{\sqrt{t}}\right).$$

Key Insight:

- If $1 - \beta_1 = O(n^{-3})$, this condition permits $b = \Omega(1)$, which allows **relatively small batchsizes**. This reveals a **Batchsize-Momentum tradeoff**.
- This convergence rate **removes the dependence of dimension d** , improving the results of prior work.

Variance-Reduction method

SVRG-style variance reduction yields **convergence to the exact full-batch max-margin solution** for **any batch size**, with or without momentum. Set $a = 1/2$. we get:

$$\gamma - \text{margin}(\mathbf{W}_t) = O\left(\frac{\frac{m^{5/2}}{1 - \beta_1}(n + \log t)}{\sqrt{t}}\right).$$

► **Key Insight:** Variance reduction **removes the limiting margin gap** $\gamma - p$ and **recover the full-batch implicit bias**, but the worst-case convergence rate contains larger batch-dependent constants, indicating that **it may converge slower**.

Extreme case: Batchsize-1

For vanilla per-sample updates, stochastic steepest descent can converge to a direction that is **fundamentally different** from **max-margin solution**.

On the orthogonal scale-skewed data $\mathbf{x}_i = \alpha_i \mathbf{e}_{y_i}$, per-sample SignSGD and Normalized-SGD make the loss converge to **0** and the solutions satisfy

$$\lim_{t \rightarrow \infty} \frac{\mathbf{W}_t}{\|\mathbf{W}_t\|_F} = \frac{\bar{\mathbf{W}}}{\|\bar{\mathbf{W}}\|_F},$$

where

$$\bar{\mathbf{W}} \triangleq \begin{cases} \sum_{i=1}^n (2\mathbf{e}_{y_i} - 1) \text{sign}(\mathbf{x}_i)^\top, & (\text{SignSGD}), \\ \sum_{i=1}^n \frac{\mathbf{e}_{y_i} - \frac{1}{K}\mathbf{1}}{\|\mathbf{e}_{y_i} - \frac{1}{K}\mathbf{1}\|_2} \frac{\mathbf{x}_i^\top}{\|\mathbf{x}_i\|_2}, & (\text{Normalized-SGD}). \end{cases}$$

► The limiting direction is governed by **label frequencies and per-sample averaging**, rather than **hard examples and max-margin geometry**.

Take away

- Without momentum, worst-case convergence and successful classification **can only be guaranteed with full-batch gradient**. (So stabilizing mechanism like momentum and Variance Reduction is important!)
- Momentum permits **smaller batches** to converge to an **approximate max-margin solution** as **full-batch implicit bias** through a batchsize-momentum trade-off.
- Variance reduction **recovers the exact full-batch implicit bias** for **any batchsize** with a slower worst-case convergence rate.
- When the batchsize is extremely small, **the max-margin behavior may collapse**. it can converge to a **fundamentally different solution** from its full-batch implicit bias.

AD: Jichu Li (4rd Undergrad) is seeking a CS/Stat/IEOR PhD position in the U.S. starting 2027fall (ML Theory & Optimization). See his work and experience at [JichuLi.github.io](https://github.com/JichuLi) ! Please feel free to reach out via lijichu52@gmail.com !